



Impact of anthropogenic climate change on Antarctic dense shelf water in a high-resolution Community Earth System Model (CESM1.3)

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Received: 17 July 2024 / Accepted: 27 August 2025

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Abstract

Dense shelf water (DSW) overflows around Antarctica leads to the formation of antarctic bottom water, thereby driving the lower cell of global overturning circulation. Some previous ocean modelling studies simulate the effects of future Antarctic glacial meltwater fluxes by adding prescribed freshwater runoff flux from the Antarctic continent. They conclude that increased meltwater is the key concern for a weakening of DSW overflows and may be responsible for a collapse of the lower cell circulation by the mid-twenty-first century. In this study, we analyze the response of DSW overflows to anthropogenic climate change in the high-resolution (0.1° ocean model) Community Earth System Model CESM1.3. In this climate model, there is no coupled dynamic ice-sheet component, and not added in prescribed glacial meltwater (although the Antarctic continent runoff is increasing under climate change forcing). Our results show that under a high-emission scenario (the Representative Concentration Pathway 8.5), the formation of DSW gradually weakens and the overflows shut down around 2040, owing to the freshening of the Antarctic shelf water. We find that over the Antarctic shelf, a reduction in sea-ice brine rejection is the dominant factor in driving the changing surface freshwater forcing (58%), while the freshwater flux from the Antarctic continent plays a secondary role (26%) and precipitation minus evaporation plays a further minor role (16%). The slowdown and eventual cessation of DSW overflow greatly weakens the lower cell of global overturning circulation. Our results highlight the critical importance of reducing the uncertainty in projected freshwater forcing **related to sea ice on the Antarctic shelf**, in addition to reducing the uncertainty in the glacial meltwater forcing. This study may help provide a more comprehensive understanding and improved **future projections of the DSW overflow in future warming climates**.

1 Introduction

Antarctic bottom water (AABW) is the coldest, densest, and most voluminous water mass of the world ocean (Orsi et al. 1999; de Lavergne et al. 2014); it fills the majority of the abyssal layers of the global ocean and plays an important role in the global meridional overturning circulation (Gordon 1986; Talley 2013). AABW is produced primarily by dense shelf water (DSW) overflow associated with coastal polynyas, although open-ocean deep convection associated with open-ocean polynyas in the Weddell Sea also rarely contributes to AABW formation (Gordon 2014; Heuzé 2021).

Around Antarctica, strong offshore winds blow newly formed sea ice away from the coast or ice shelf fronts, leading to the formation of coastal polynyas (e.g., Morales Maqueda et al. 2004; Stössel et al. 2011; Zhang et al. 2015). The open water in coastal polynyas is exposed to cold air masses (about −30 °C; Turner et al. 2009) that descend from

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the Antarctic ice sheet, which leads to vigorous new-ice production and brine rejection (Tamura et al. 2008, 2016). The consequently increased salinity at the surface eventually generates High Salinity Shelf Water (HSSW). In some locations (e.g., Ross and Weddell Seas shelf), HSSW can flow into the ice shelf cavity. While the sea water freezing point is pressure dependent, HSSW is warm enough to cause melting at the ice shelf base. This produces supercooled water, typically below $-2\text{ }^{\circ}\text{C}$ (below the surface freezing point), which is termed as the Ice Shelf Water (ISW). HSSW and ISW together are usually referred to as DSW (Silvano et al. 2023; Naughten et al. 2019). After filling the bottom of the shelf and reaching the shelf break, DSW starts spilling off the shelf. While cascading down the continental slope, DSW mixes with lighter ambient water (mostly Circumpolar Deep Water or Antarctic Surface Water in some cases) to finally form AABW. This DSW overflow process is the main source of AABW and occurs in four main regions around Antarctica: the Weddell Sea, the Ross Sea, Prydz Bay (Cape Darnley), and the Adélie Coast (Orsi et al. 1999; Purkey et al. 2018).

Open-ocean polynyas form offshore and mostly in the Weddell Sea (e.g., Morales Maqueda et al. 2004; Heuzé 2021). They tend to form sporadically at distinct locations that are preconditioned by weak stratification (e.g., Gordon 2014; de Lavergne et al. 2014; Stössel et al. 2015). Polynyas forming over and around the Maud Rise seamount have been termed “Maud Rise Polynyas” (MRPs), while the term “Weddell Polynya” (WP) has been used to describe the large polynyas that emerge intermittently on inter-decadal timescales in the central Weddell Sea (Gordon and Huber 1990; Campbell et al. 2019; Kurtakoti et al. 2018, 2021; Diao et al. 2022). Surface cooling can make the anomalously salty surface water dense enough to sink, and be replaced by the warmer and saltier Weddell Deep Water (WDW) from below, thereby sustaining the deep convection (e.g., Dufour et al. 2017) and maintaining ice-free (polynyas) conditions (Gordon 1978). Open-ocean deep convection can reach as deep as 3000 m (Gordon et al. 2014), and can thus provide a secondary mechanism of AABW production. The “WP” (Gordon et al. 2007) was first observed in the winter of 1974 when the first satellite passive-microwave data became available, and it reoccurred over the next two consecutive winters (1975–1976) (Carsey 1980; Zwally et al. 1983). Since then, such large polynyas have not been observed again, probably owing to a fresher, more stratified Weddell Sea (de Lavergne et al. 2014). During the past four decades, AABW thus formed almost exclusively via DSW overflow (Gordon 2014). In this study, we mainly focus on the DSW overflow and its future projection.

In addition, it is still challenging to conduct direct measurements of AABW formation rate, as the DSW overflow

spatial scales are small. Thus, the current available data sets are mainly limited to hydrographic section observations in summer, or some sporadic moorings or Argo floats observations (Gordon et al. 2020; Castagno et al. 2019; Foppert et al. 2021). Tracers, like the concentrations of chlorofluorocarbons, can also be used to estimate the AABW formation and transport rate (Orsi et al. 1999; Purkey et al. 2018). Orsi et al. (1999; 2002) estimated that all Antarctica AABW export of about $8.1 \pm 2.6\text{ Sv}$ based on the tracer method. Furthermore, the estimated DSW export flux was estimated to be about 3.4 to 4.9 Sv in the Weddell Sea (Orsi et al. 1999; Meredith et al. 2015), and to be about 1.7 Sv (Gordon et al. 2009) or about $1.95 \pm 1.85\text{ Sv}$ (Whitworth and Orsi 2006) in the Ross Sea.

The ability of an ocean model to adequately depict AABW formation processes is crucial for a realistic simulation and projection of changes of the global bottom water properties and hence the global meridional overturning circulation. However, the models that participated Coupled Model Intercomparison Project Phase 5 and Phase 6 (CMIP5 and CMIP6) can still not properly represent DSW overflows and often have excessive open ocean deep convection processes due to their relatively low-resolution (about 1° ; Heuzé et al. 2013; Heuzé 2021). Recently, high-resolution (about 0.1°) global ocean-sea-ice model simulations revealed fundamental improvements in the simulation of DSW overflow process due to their better representation of the steep Antarctic shelf bathymetry and the Antarctic shelf sea dynamics (Morrison et al. 2020, 2023; Moorman et al. 2020; Schmidt et al. 2023; Li et al. 2023; Jeong et al. 2023). For example, based on a high-resolution global ocean-sea-ice model, Morrison et al. (2023) found that the enhanced offshore meridional wind is the main driver of the enhancement of both DSW production and the abyssal overturning circulation. This is due to that the meridional wind increases the northward export of sea ice, and then increases the sea ice formation and hence brine rejection. In contrast, the zonal easterly winds play a little role in this DSW production process. However, based on a similar model, Schmidt et al. (2023) suggested that easterly winds play an important role in the DSW production, with the weaker easterly winds reducing the sea ice import into the DSW formation regions. Furthermore, based on a transient simulation using the same high-resolution global ocean-sea-ice model, Li et al. (2023) showed that artificially added ice-sheet meltwater input around Antarctica is the dominant driver for the slowdown of the DSW overflow and also for the contraction of AABW, while the projected winds and thermal forcing have very small impacts. In addition, the reduction of DSW overflow results in the warming and ageing of the deep ocean under their climatic forcing. In Li et al. (2023), the ocean-sea ice model was uncoupled with the atmospheric component, and driven by projected

atmospheric forcing derived from CMIP models under a high-emissions scenario and by the increased glacial meltwater derived from glacial ice models. By analyzing the results from a high-resolution fully coupled model (CESM), Jeong et al. (2023) showed that the warming is the main driver for the slowdown of the DSW overflow owing to the decline of sea ice production, and they suggested that the increased precipitation also drives the salinity decrease on the Antarctica shelf (10–20 Gt/year (0.00032–0.00064 Sv) for precipitation vs 100 Gt/year (0.0032 Sv) for ice-shelf melting in their estimate, and The Ice Sheet Mass Balance Inter-Comparison Exercise team (IMBIE) observational results showed that Antarctic ice sheets have lost ice mass at an accelerating rate in recent years. Between 1992 and 2011, the ice mass loss was 76 ± 59 Gt/year (0.0024 ± 0.0019 Sv), and between 2012 and 2017, it was 219 ± 43 Gt/year (0.0069 ± 0.0014 Sv) (IMBIE team 2018)).

In this research, by analyzing a transient simulation from a high-resolution fully coupled Community Earth System Model version 1.3 (CESM1.3) under a high-emission scenario, we find that although no artificially increased meltwater from the Antarctic ice sheet is included, the DSW overflow also ceases. This result indicates that it is interesting to find what drive this DSW overflow cessation in this particular model, which, we believe, may work together with the ice-sheet meltwater mechanisms (e.g., Li et al. 2023) to weaken the DSW overflow. Furthermore, we do a detailed analysis of the total freshwater flux budget on the Antarctica shelf in the CESM1.3 POP model to disentangle the roles played by brine rejection and precipitation in driving the DSW overflow cessation. Our results highlight the importance of reducing the uncertainty in the projected change in freshwater forcing on the Antarctic shelf.

This paper is organized as follows. Section 2 gives a brief description of the model, the simulation settings, and the analysis methods. Section 3 describes the main results, with a focus on the effect of the anticipated future anthropogenic forcing on (a) Antarctic shelf water properties, (b) DSW overflow evolution, and (c) abyssal global overturning circulation response, (d) the characteristic vertical sections on the Ross Sea shelf and the Weddell Sea shelf. Why the DSW overflow ceases in a warming climate in this model is then discussed in (e). We finally provide a summary and concluding remarks in Sect. 4.

2 Model and methodology

2.1 CESM1.3

The model results are from the fully coupled Community Earth System Model version 1.3 (CESM1.3) developed by

the National Center for Atmospheric Research (NCAR) in collaboration with the broader climate modeling community (Small et al. 2014; Meehl et al. 2019; Chang et al. 2020). The CESM1.3 consists of Community Atmosphere Model version 5 (CAM5; Neale et al. 2010; Dennis et al. 2012), the Parallel Ocean Program version 2 (POP2; Danabasoglu et al. 2012; Smith et al. 2010), the Community Ice Code version 4 (CICE4; Hunke et al. 2008), and the Community Land Model version 4 (CLM4; Lawrence et al. 2011). Our high-resolution simulations here used 0.1° for POP2 and CICE4 and 0.25° for CAM5 and CLM4 resolution. However, CESM1.3 has no interactive ice shelf and ice sheet components. Instead, the snow/ice discharge from Antarctica is treated with a snow-capping parameterization. The snow-capping parameterization is activated whenever the snow depth in a grid cell exceeds a prescribed maximum value, set to 1 m in this study. Upon activation, the parameterization redistributes the excessive snow (ice flux) into the ocean via a nearest-neighbor algorithm, spreading it using a Gaussian distribution over a radius of 300 km. If the ocean is warm enough, this ice flux is melted in the ocean immediately. While ice discharge is simulated as "capping" of multi-year snow accumulation, the meltwater (liquid) runoff is simulated according to a surface energy budget scheme to calculate snow and ice surface melt (Danabasoglu et al. 2012; Jeong et al. 2023). We do not use salinity restoring in these fully coupled CESM1.3 simulations. No overflow parameterization is employed in this study (Danabasoglu et al. 2012). The tidal effects are not included (Chang et al. 2020).

2.2 High-resolution preindustrial control simulation (HR-PI)

The experimental design of HR-PI follows the protocol of the CMIP phase 5 (CMIP5) experiments (Taylor et al. 2012). The POP2 initial conditions were derived from the climatological January-mean World Ocean Atlas (WOA) potential temperature and salinity (Locarnini et al. 2018; Zweng et al. 2019), while CICE4, CAM5, and CLM4 were initialized by using restart files of previous simulations (Chang et al. 2020). Notably, no mesoscale and sub-mesoscale eddy transport parameterizations were used. The radiative forcing in HR-PI was set to conditions in 1850 and kept constant throughout the entire 500-year simulation.

2.3 High-resolution historical and future transient simulation (HR-TN)

The HR-TN simulation was branched from the year 250 of HR-PI. The historical forcing from 1850 to 2005 was used for HR-HIST, and then the Representative Concentration

Pathway 8.5 (RCP8.5) forcing from 2006 to 2099 was used for HR-TN (Chang et al. 2020). Furthermore, the ozone concentrations were calculated using a high-top coupled chemistry-climate model (Marsh et al. 2013) considering specified ozone depleting substances. In this study, we consider both the atmospheric carbon dioxide concentration change and the stratospheric ozone concentration change as the “anthropogenic forcing”.

2.4 Analysis methods

We used the Surface Water Mass Transformation (SWMT) framework following Newsom et al. (2016) and Yeager et al. (2021) to estimate the DSW formation rate on the Antarctic shelves. The surface water mass transformation rate is defined as the volume flux into a particular potential density bin due to the surface heat (Q) and freshwater forcing (F) as follows:

$$f = -\frac{\alpha}{C_P}Q - \beta \frac{S}{1-S}F,$$

where f is the surface density flux, α is thermal expansion coefficient, β is the haline contraction coefficient, C_P is the specific heat capacity of sea water, and S is the sea surface salinity. Finally, f is integrated over the surface density outcrop regions (A_ρ) to yield the SWMT as a function of density (ρ) (Yeager et al. 2021):

$$SWMT_\rho = \frac{1}{\Delta\rho} \iint f dA_\rho.$$

In addition, we also estimated the DSW overflow export across the 1000-m isobath, which was calculated offline using monthly averaged output and was conservatively binned into σ_1 isopycnal. Then we cumulatively summed the transport in potential density space from the bottom upwards across all density layers. The offshore transport of DSW overflow bigger than the density of $\sigma_1 = 32.57 \text{ kg/m}^3$ (the same definition as in Li et al. 2023) was referred as the DSW overflow export throughout this study.

Furthermore, we do a freshwater flux budget analyzation on the Antarctica shelf. In CESM1.3, the POP ocean component model total surface freshwater flux can be broken into seven components as follows:

$$SFW_f - Q_f/Q_{term} = (P_f + E_f) + (R_f + I_f) + (MELT_f + SALT_f/S_{term} - Q_f/Q_{term}).$$

$SFW_f - Q_f/Q_{term}$ represents the ocean total freshwater flux. The Q_f (noted in the CESM1.3 model as QFLUX) term isn't included in SFW_f (noted in the model as SFWF) term because it's not considered a flux exchanged between

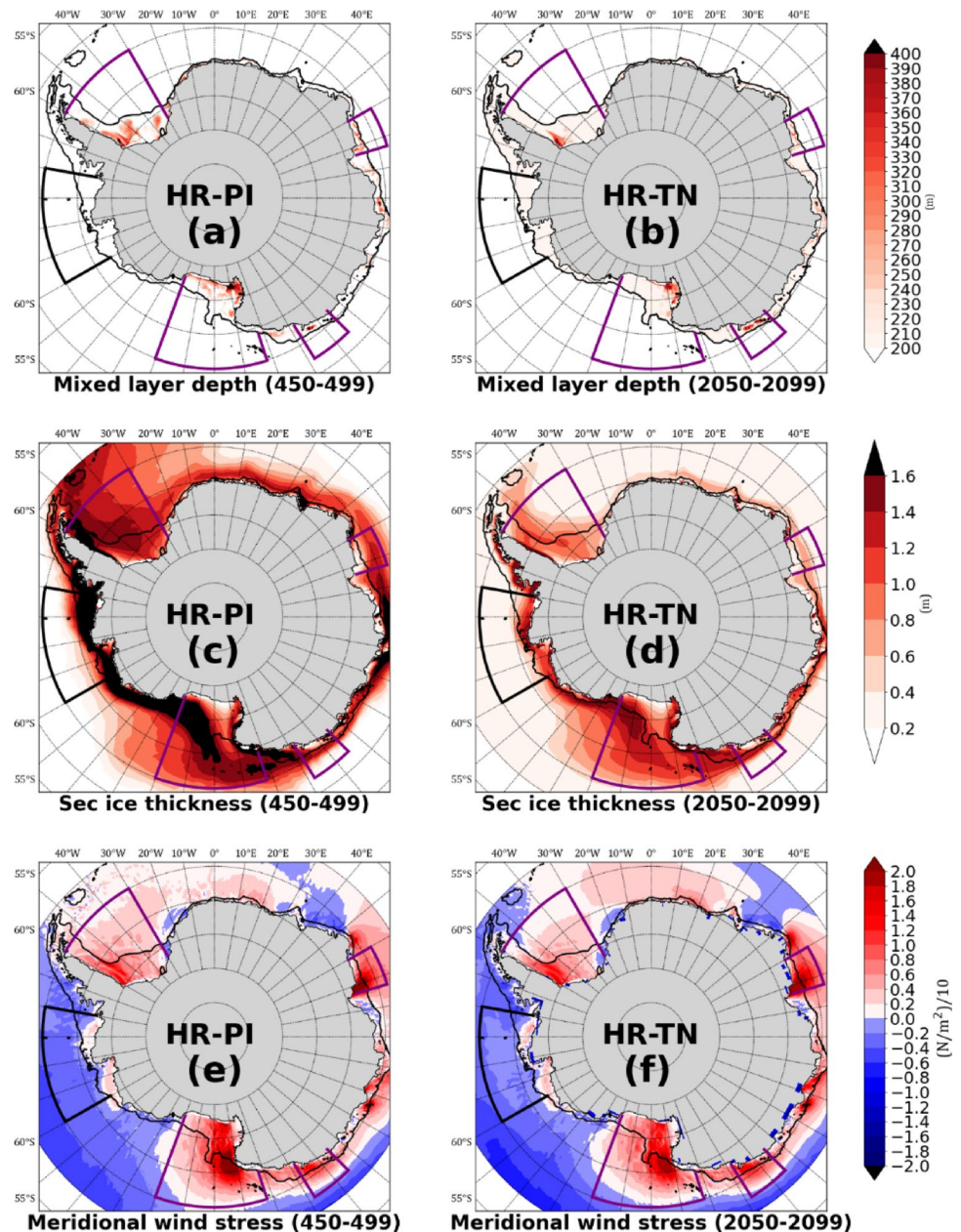
model components, instead, it is the freshwater flux associated with frazil ice formation which occurs within the ocean model. Q_{term} is the latent heat of fusion term (noted in the model as latent_heat_fusion/10000), which is needed for correct units here. $(P_f + E_f)$ represents the freshwater flux exchange between atmosphere and ocean (noted in model as PREC_F+EVAP_F term). $(R_f + I_f)$ represents the freshwater flux exchange between Antarctica continent and ocean. R_f represents the liquid water runoff (noted in the model as ROFF_F term), while I_f represents the snow/ice runoff from the Antarctica continent (note in the model as IOFF_F term). $MELT_f$ (noted in the model as MELT_F term) represents any freshwater flux computed by the sea ice model and sent to the ocean (associated with sea ice melt/growth), while $SALT_f$ (noted in the model as SALT_F term) represents any SALT flux computed by the sea ice model and sent to the ocean (associated with the fact that ice melt/growth requires that salt be added/removed from the ocean because ice has a constant salinity of ~4 psu). This salt flux must of course be converted to appropriate units by using the S_{term} (noted in the model as sflux_factor/salinity_factor) term in the model. $(MELT_f + SALT_f/S_{term} - Q_f/Q_{term})$ Combined to be considered to be the freshwater flux that associated with the sea-ice brine rejection and melt effect. Finally, the salinity restoring is not used in these fully coupled CESM1.3 simulations.

3 Results

3.1 Anthropogenic forcing influence on Antarctica shelf water properties

Since the DSW formation and overflow are closely related to the Antarctic coastal polynyas, we first examine the fidelity of the coastal polynyas simulated in HR-PI. The simulated mixed-layer depth (Fig. 1a) and sea ice thickness (Fig. 1c) patterns in September suggest that HR-PI fairly captured the observed distributions of the coastal polynyas around Antarctica in winter. In these coastal polynyas, sea ice remains very thin and drifts away from the coasts or ice shelf fronts driven by offshore katabatic winds (Fig. 1e; see also Stössel et al. 2011 and Morrison et al. 2023), such that new ice is continuously being produced. In these “ice factories” (Tamura et al. 2016), salt is continuously added to the surface water, which eventually makes the surface water in these coastal polynyas dense enough to sink. The yearly-mean surface water mass transformation across $\sigma_1 = 32.3 \text{ kg/m}^3$ (Fig. 2a, b) indicates that the DSW transformation mainly happens on the western Ross Sea shelf, the Adélie Coast, the western Weddell Sea shelf, and the Prydz Bay

Fig. 1 **a** Mixed layer depth (defined as the shallowest depth where the local buoyancy gradient equals the largest buoyancy gradient of the total water column) in September on the Antarctic continental shelf averaged over years 450–499 simulated by HR-PI, and **b** over years 2050–2099 simulated by HR-TN. **(c), (d)** Similar to **(a)** and **(b)**, but for sea-ice thickness (with thin ice along the coast indicating coastal polynyas). **(e)** and **(f)** Similar to **(a)** and **(b)**, but for meridional wind stress (positive northward). The characteristic *dense shelf* regime is normally found across the shelf break within the purple boxes, while the *warm shelf* regime is within the black box. The shelf regions outside these boxes are mostly classified as *fresh shelf* regimes. The black contour line is the 1000 m isobath



(Cape Darnley) region. This pattern matches well with previous observational results and other model results (Morrison et al. 2020; Li et al. 2023; Whitworth and Orsi 2006; Gordon et al. 1998; Williams et al. 2010; Ohshima et al. 2013). Comparing the right panels with the left panels in Fig. 1, we can see that the anthropogenic forcing has a big impact on the sea ice thickness and the mixed layer depth, and the offshore wind strengths also reduce to some extent. Furthermore, Fig. 2a–f clearly show the change in the Antarctica shelf surface water mass transformation under the climate change forcing from year 2010 to 2060. After about year 2040, no surface water can be transformed into DSW that is denser than $\sigma_1 = 32.3 \text{ kg/m}^3$ anymore.

Furthermore, we examine the Antarctic shelf bottom water properties. Figure 3a shows that salty water (salinity $> 34.85 \text{ psu}$) is produced in HR-PI at the western Ross Sea shelf bottom, which matches well with the observation results of Orsi and Wiederwohl (2009; $> 34.82 \text{ psu}$; their Fig. 8c) and Whitworth et al. (1998; $> 34.9 \text{ psu}$; their Fig. 11b). At the bottom of western Weddell Sea shelf, the Adélie Coast, and the Prydz Bay (Cape Darnley) region, HR-PI also produces high salinity shelf waters with salinity of about 34.7 psu (Fig. 3a), consistent with the observation results of Whitworth et al., (1998). In concert with the high shelf salinity, HR-PI also reproduces the observed low temperature ($< -1.7 \text{ }^\circ\text{C}$; see Fig. 11a of Whitworth et al. 1998)

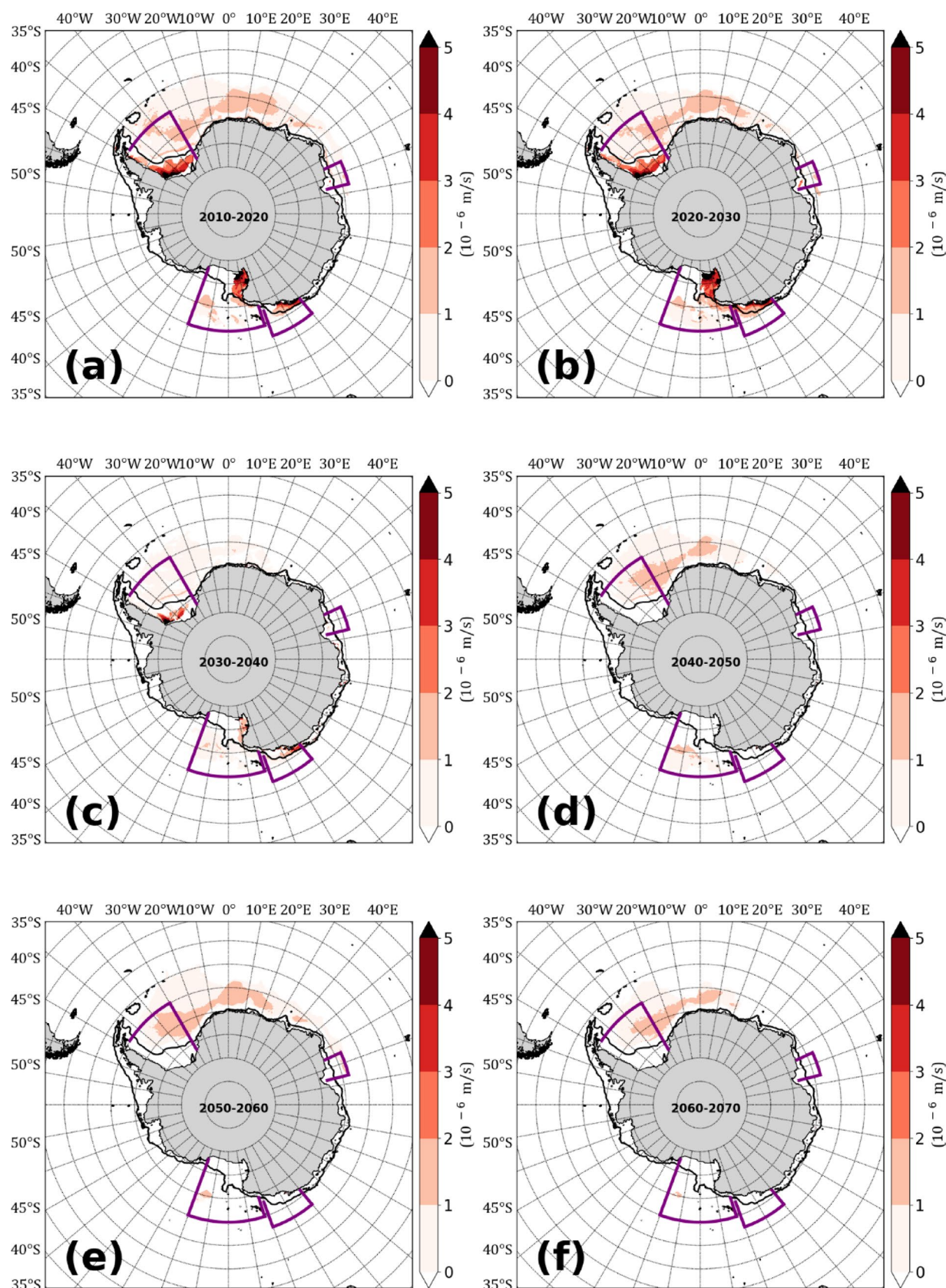


Fig. 2 a Decadal-mean winter (May to October) surface water mass transformation across $\sigma_1 = 32.3 \text{ kg/m}^3$ in year a–f 2010–2020, 2020–2030, 2030–2040, 2040–2050, 2050–2060, 2060–2070 in HR-TN, respectively. The four purple boxes are the four observed DSW overflow sites

on the Ross Sea and Weddell Sea shelf (Fig. 3c). Thus, DSW with high salinity and low temperature is produced (Fig. 3e), with the densest water located in the western Ross Sea ($\sigma_1 > 32.57 \text{ kg/m}^3$; consistent with Legg et al. 2009 and Morrison et al. 2020). Note that the shelf water density there is even higher than in the abyssal ocean; these are the sites where we would expect DSW overflow to occur. The DSW on the western Ross Sea shelf cascades down to the abyssal ocean, and is then transported eastward to the central Ross Sea (Fig. 3e) (also in the bottom water age plot; Fig. S1). In the Weddell Sea, on the other hand, although the westward DSW transport pathway is quite clear on the shelf, the DSW downslope cascading pathway is not that obvious (this may be due to the influence of the open-ocean deep convection that also occurs in the central Weddell Sea in our experiments). Comparing the right panels and the left panels in Fig. 3, we can see that the anthropogenic forcing in HR-TN makes all the bottom water on the Antarctica shelf fresher, warmer, and less dense. Especially on the western Ross Sea shelf and on the western Weddell Sea shelf, when the shelf bottom water potential density becomes smaller than that of the abyssal ocean, the DSW overflow process will finally stop (right panels in Fig. 3).

3.2 Anthropogenic forcing influence on DSW overflow

We now explore the detailed changes of the DSW overflow process. Observations show that the entire water column salinity in the Ross Sea shelf region has shifted toward lower values since the 1960s (Jacobs et al. 2002, their Fig. 2; Jacobs and Giulivi 2010). To follow up on this in our simulations, we examine the time series of the whole water column salinity on both the Ross Sea and Weddell Sea shelves (Fig. 4). First, as expected, the Ross Sea shelf salinity is a little higher (about 0.1 psu) than that of the Weddell Sea in HR-PI. During the historical climate forcing from year 1850–2005 (before the black dashed line), there is a weak negative trend from 1960 to 2000 or from 1965 to 2005 of HR-TN, with the weak negative trend of about -0.03 psu/decade similar to the observed result (Jacobs et al. 2002; Jacobs and Giulivi 2010; Schmidt et al. 2014). In contrast, after the onset of the RCP8.5 climate forcing from 2006 to 2100 (after the black dashed line), the salinity decline is much faster (-0.07 psu/decade in the Weddell Sea and -0.09 psu/decade in the Ross Sea).

Furthermore, we also calculated the export of DSW from the continental shelf across the 1000-m isobath. Figure 5 shows the 5-year running mean time series of DSW export. The HR-PI experiment simulates DSW export of about 6.1–17.5 Sv across the 1000-m isobath (the strong decadal fluctuations are related to open-ocean deep convection, this

is ongoing research now). In contrast, the HR-TN experiment shows that the DSW export is considerably weakening in response to the projected changes in anthropogenic forcing. Note that the observational DSW export during year 1980–2000 was estimated to be about $8.1 \pm 2.6 \text{ Sv}$ (Orsi et al. 2002), which is in the range of our simulated value of about 7.5–13 Sv during the same period (Fig. 5). As the CO_2 concentration continues to increase, especially under the RCP8.5 climate forcing from year 2006–2100, the DSW export eventually becomes vanished around year 2035. The projected decline of DSW export drives a warming and freshening trend in the abyssal Southern Ocean (Fig. 3c, d), reminiscent in structure to recently observed bottom water trends.

To investigate the detailed cessation process of DSW overflow, we zoom in and focus on the 2010–2090 period, when the model simulates the transition process from the high DSW production rate to no DSW overflow features (Fig. 6). Under the RCP8.5 climate forcing, the surface water mass transformation shows a broadly linear decreasing trend for all density bins (Fig. 6a). After about year 2040, there is no more surface water that can be transformed into $\sigma_1 > 32.57 \text{ kg/m}^3$ density layers on the Antarctic shelf (Fig. 6a). The DSW export across 1000 m isobath has quite similar spatial patterns (Fig. 6b). However, due to the entrainment in its transport process, DSW around 1000 m isobath normally has a smaller density than at the DSW formation site for all density bins (Fig. 6b vs Fig. 6a). The DSW overflow across 1000 m isobath with $\sigma_1 > 32.57 \text{ kg/m}^3$ becomes vanished around year 2040.

To investigate the relative roles of controlling factors, we decompose the surface water mass transformation into the thermal and haline parts induced by heat flux and freshwater flux (Fig. 7). We then further decompose the freshwater flux into three components: precipitation minus evaporation on the Antarctic shelf, snow/ice runoff and liquid water runoff from the Antarctic continent, and sea-ice brine rejection and melt on the Antarctic shelf. The result indicates that the formation of DSW is dominated by sea-ice brine rejection effect ($\sigma_1 > 32.57 \text{ kg/m}^3$, the dark-red horizontal line), with the snow/ice runoff and liquid water runoff from the Antarctic continent playing an opposite role to make the shelf water less dense. In addition, the effects of heat flux and precipitation minus evaporation are very minor. This is consistent with other high-resolution global simulation results (Morrison et al. 2020) and an observation-based estimate (Pellichero et al. 2018). Furthermore, the dominant effect of sea ice brine rejection holds for all the transition period under the climate change forcing (Fig. 7a–f). However, after year 2040, under the climate change forcing, the brine rejection effect becomes so weak (Fig. 7d–f) that no DSW ($\sigma_1 > 32.57 \text{ kg/m}^3$) on the Antarctic shelf is produced anymore.

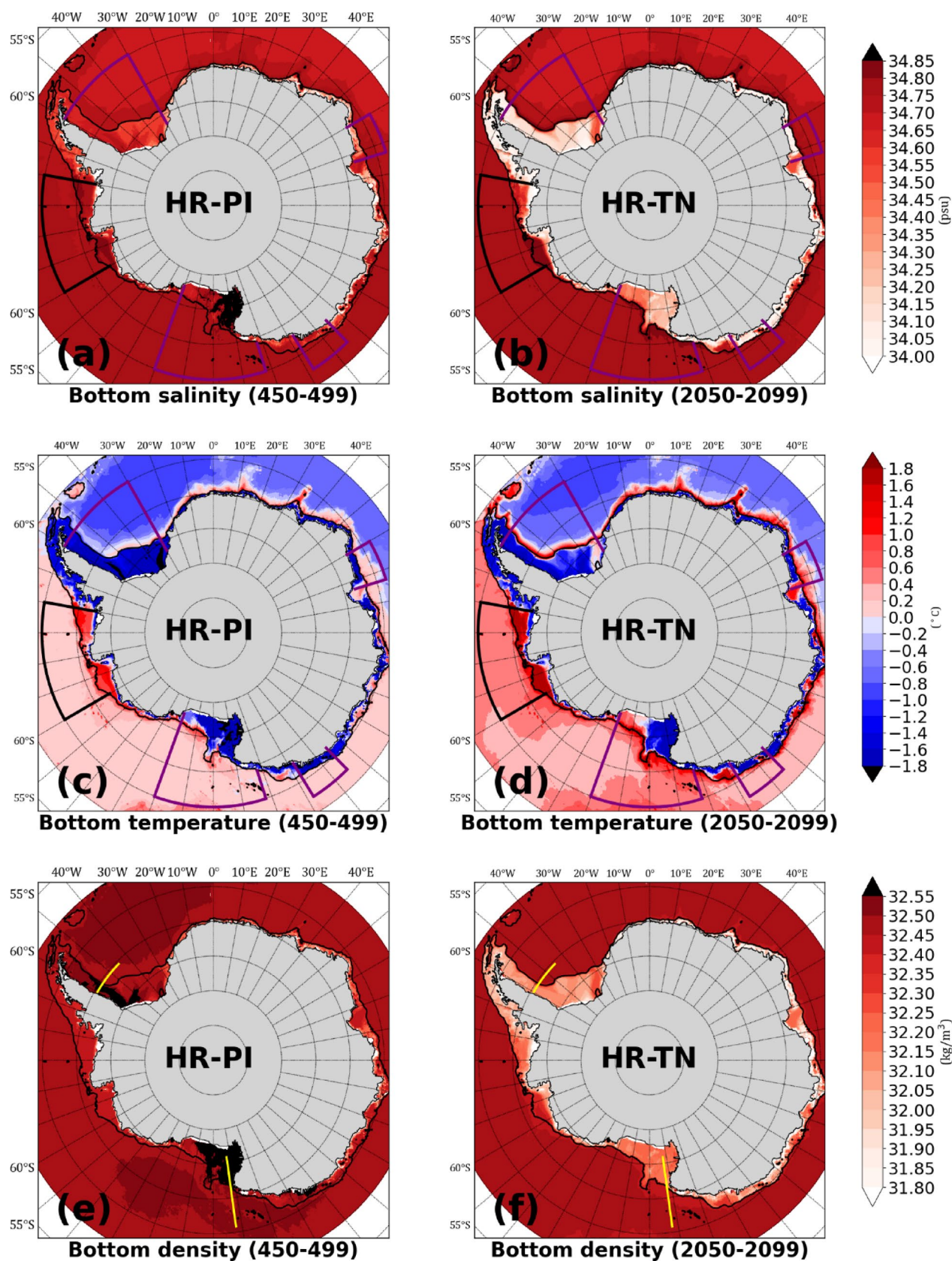


Fig. 3 **a** Salinity in September in the bottom layer averaged over years 450–499 of HR-PI, and **b** over years 2050–2099 of HR-TN. **(c)** and **(d)** Similar to **(a)** and **(b)**, but for potential temperature (1000 m reference depth). **(e)** and **(f)** Similar to **(a)** and **(b)**, but for potential density σ_1 . The meanings of the boxes in **(a)** through **(d)** and the black contour line are the same as in Fig. 1. The two yellow lines in **(e)** and **(f)** indicate the horizontal-vertical sections displayed in Figs. 10 and 11

3.3 Global overturning circulation response

The HR-TN experiments show that the global overturning circulation is expected to weaken in response to projected climate change forcing (Fig. 8). As shown in Fig. 8a–c, the global overturning circulation includes two meridional overturning cells in density space: namely, the clockwise upper cell (shown in red colors), and the anti-clockwise lower cell (shown in blue colors). In this study, we mainly focus on the responses of the lower cell to the change of DSW overflow. The maximum strength of the lower overturning cell is about 29 Sv in year 2010, which is similar to the results estimated from observations (Lumpkin and Speer 2007; Talley 2013). As the climate gets warmer over time, the DSW overflow ceases around year 2040 (Fig. 6), and the lower cell starts to separate into two parts in density space: the lighter part's center is in the range of $\sigma_2 = 36.8\text{--}37.0 \text{ kg/m}^3$, while the denser part's center is at about $\sigma_2 = 37.25 \text{ kg/m}^3$ (Fig. 8d–i). Furthermore, the lighter part shows a roughly linear decreasing trend under the climate change forcing (Fig. 8d–i), while the denser part shows a decadal fluctuation (especially around year 2060; may be caused by strong open-ocean deep convection) before it eventually becomes very weak. Until year 2100, the overall strength of the lower overturning cell decreases to about 5 Sv: with the lighter part's strength of about 3 Sv and the denser part's strength of about 2 Sv.

Applying the surface water mass transformation framework only on the shelf water (<1000 m only) and only on the open ocean water (>1000 m only) separately, we can see that after year 2040, the dense water with $\sigma_2 > 37.25 \text{ kg/m}^3$ can only be produced from the open-ocean deep convection (Fig. 9b), and not from DSW overflow anymore (Fig. 9c). Thus, the reason for the lower cell to separate into two parts is that the lighter part of the lower cell is driven by the DSW overflow, which is further dominated by the brine rejection effect (Fig. 7). The roughly linear decreasing trend of the brine rejection effect and the Antarctic shelf salinity (Figs. 4 and 7) makes the DSW overflow weaker and weaker over time. Finally, around year 2040, the shelf water is not dense enough to cascade down to the bottom of the abyssal ocean, and the lower cell starts to separate into two parts (Fig. 8d, e). After year 2040, the brine rejection effect continues to become weaker, which further decreases the shelf water density and makes the lighter part's central position to move upward continuously in density space (Fig. 8e–j).

On the other hand, the denser part of the lower cell is driven by the open-ocean deep convection, which fluctuates on decadal scale. The unrealistic open-ocean deep convection in the Weddell Sea (reference to Diao et al. 2022 to more detailed discussion about this topic) can still produce dense water with $\sigma_2 > 37.25 \text{ kg/m}^3$, even under the high emission climate change forcing after year 2050 (Fig. 9a, b; see also in Fig. 2f). Furthermore, the surface water transformation of the open-ocean deep convection is dominated by temperature (instead of brine rejection effect on the shelf), and finally contribute to the fluctuations in the denser part of the lower cell in global overturning circulation (Fig. 8).

3.4 Meridional changes

Vertical cross-sections can provide some more intuitive insight into the physical mechanisms regulating DSW overflow in HR-PI and HR-TN simulations. Following some previous studies (Thompson et al. 2018; Gordon et al. 2009, 2020), we select two distinct typical vertical sections to investigate how the anthropogenic forcing influences the cross-slope dynamics (their locations are marked in Fig. 3e).

Figure 10 shows the water mass distributions resulting from the DSW overflow process along the Ross Sea shelf section (172°E, near the Drygalski Trough; similar to Fig. 1 of Gordon et al. (2009), its location is shown in Fig. 3e). The relatively cold (<0 °C) and fresh (about 34.76 psu) DSW along the continental slope is well represented in HR-PI (Fig. 10a, c). The properties also match with the observational results of Gordon et al. (2009; their Fig. 6). Furthermore, in HR-PI, we can see that the DSW overflow is a salinity-controlled phenomenon (Fig. 10c). The High Salinity Shelf Water produced in the Ross Sea coastal polynyas increases the water density. The resulting DSW accumulates on the Ross Sea shelf (with potential density $\sigma_1 > 32.57 \text{ kg/m}^3$ and salinity >34.82 psu). After the DSW fills the bottom layers of the shelf, it begins to overflow across the shelf break. As the DSW cascades down the slope, it mixes with the adjacent water masses (mostly Circumpolar Deep Water), reducing the density of the resulted water mass, i.e., AABW (about 32.49 kg/m^3). In HR-TN, the shelf water becomes much fresher in the second half of the twenty-first century, which eventually leads to the cessation of DSW production and hence DSW overflow (Fig. 10b, d). Furthermore, the reduction of the DSW production originates at the surface (in coastal polynyas) where the salinity (density) of the surface water is being reduced to the point that it cannot sink to the bottom of the shelf. With the lack of high-density “source” water production, the DSW overflow ceases during the last 50 years of HR-TN (the right panels in Fig. 10). The cessation of DSW overflow leads to the buildup of a deep heat and salt reservoir along the continental slope

Fig. 4 Time series of water-column mean salinity in September of HR-PI (black) and HR-TN (red), averaged over **a** the Weddell Sea shelf region, and **b** the Ross Sea shelf region. The dashed line indicates year 2006 of HR-TN, the start year of the RCP8.5 forcing

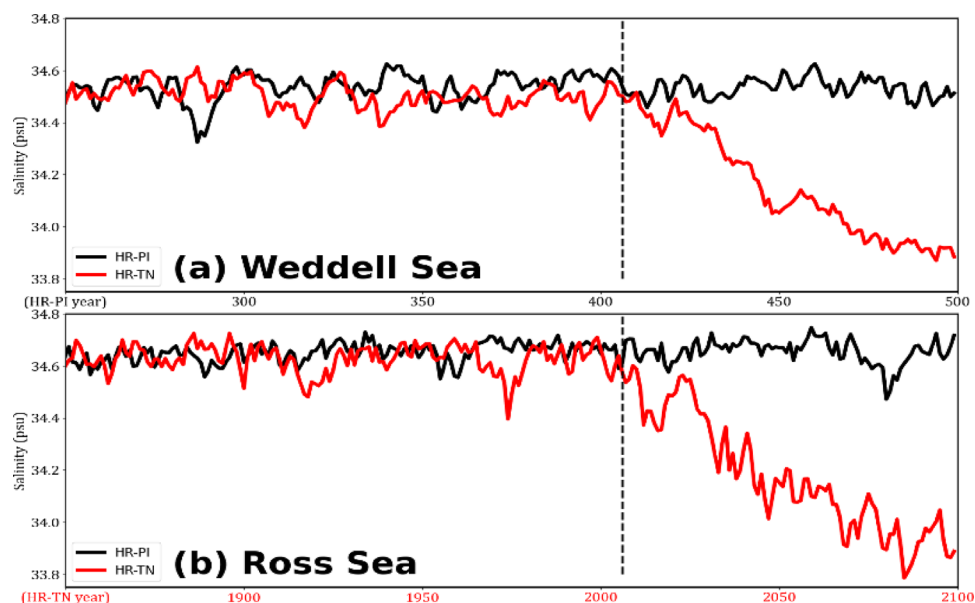
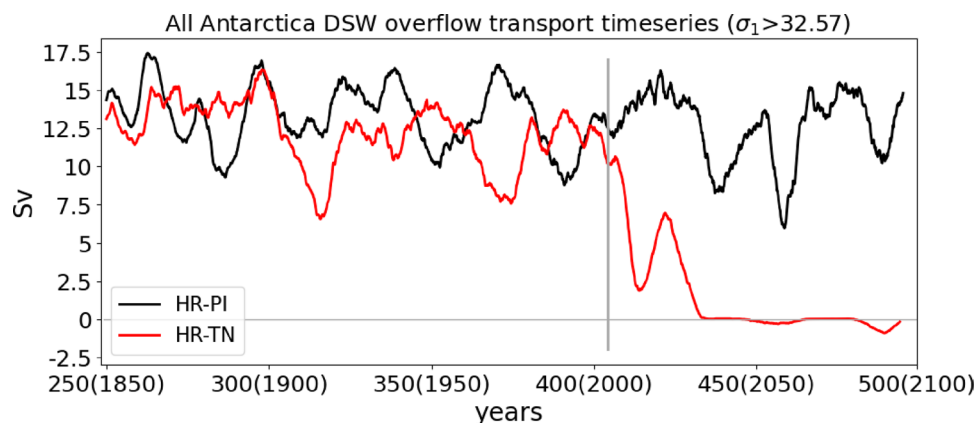


Fig. 5 Time series of the 5-year running mean time series of circumpolar DSW overflow export across 1000-m isobath for $\sigma_1 > 32.57 \text{ kg/m}^3$ in HR-PI (black) and HR-TN (red). The dashed line indicates year 2006 of HR-TN, the start year of the RCP8.5 forcing



(right panels of Fig. 10). The Circumpolar Deep Water significantly warms and moves closer to the shelf in HR-TN.

Figure 11 shows the DSW overflow process in the selected dense shelf section of the Weddell Sea (67.5°S ; matched with the observational section in Fig. 1 of Gordon et al. (2020)). The observed “V” shaped isopycnals (e.g., Whitworth et al. 1998) are captured very well in HR-PI (Fig. 11a), which indicates that the HR simulations are capable of reproducing DSW overflow in the specific regions where it has been observed. The temperature pattern in HR-PI agrees well with the observational result of Gordon et al. (2020), with the low temperature water (less than -1°C) below 1000-m depth (Fig. 11a). The HR-PI salinity also agrees well with the result of Gordon et al. (2020), with salinity larger than 34.64 psu on the Weddell Sea shelf. Similar to the section in the Ross Sea, the water on the Weddell Sea shelf also becomes much fresher during the last 50 years of HR-TN, which eventually leads to the cessation of DSW overflow there (the right panels in Fig. 11). A deep heat and salt reservoir along the continental slope also

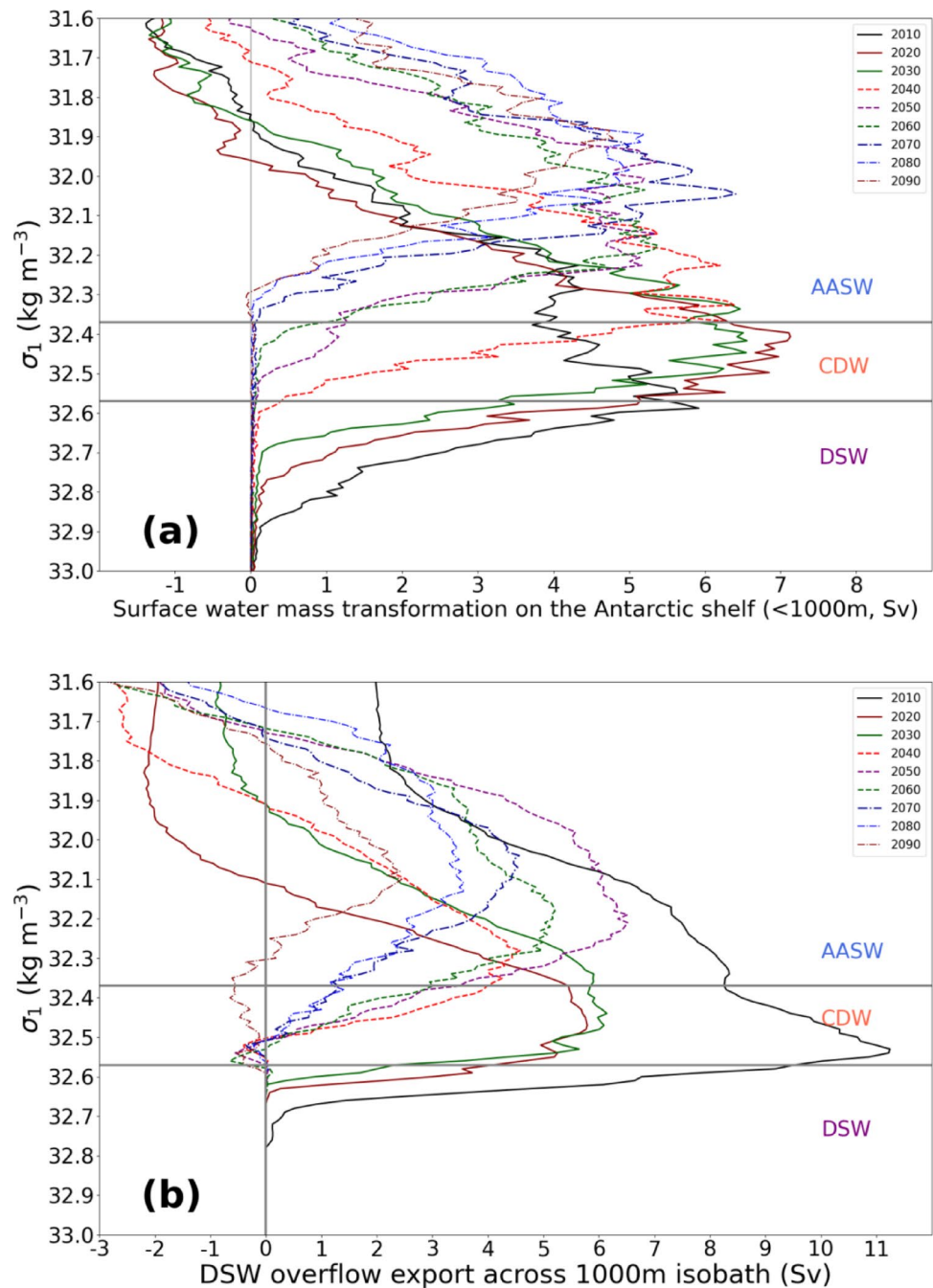
builds up, due to the cessation of DSW overflow. The Circumpolar Deep Water also significantly warms and moves closer to the shelf in HR-TN in this section.

3.5 Causes for DSW overflow collapse

Both observational and modeling studies have suggested a number of factors that can make the Antarctic shelf water fresher, including: (1) reduced sea ice production and the associated brine rejection effect, (2) increased precipitation minus evaporation, and (3) increased melting of the Antarctic ice sheet (e.g., Shepherd et al. 2018; Snow et al. 2016; Jacobs et al. 2002; Jacobs and Giulivi 2010; Lockwood et al. 2021).

Regarding factor (3), Li et al. (2023) imposed a linear increase of freshwater input into the Antarctic shelf to mimic the ice-sheet melting process in a high-resolution global ocean-sea-ice model. They concluded that such an ice-sheet melting process is the most dominant factor that drives the weakening of DSW overflow. In contrast, projected changes

Fig. 6 **a** Surface water mass transformation on the Antarctica shelf (isobath < 1000 m), cumulatively integrated up-ward through density space (yearly-mean data). No more DSW ($\sigma_1 > 32.57 \text{ kg/m}^3$) is produced after year 2040. **b** Net offshore DSW overflow transport across the 1000-m isobath, cumulatively integrated up-ward through density space (yearly-mean data). The DSW overflow stops between years 2030–2040. The horizontal lines and colored labels show water mass definitions

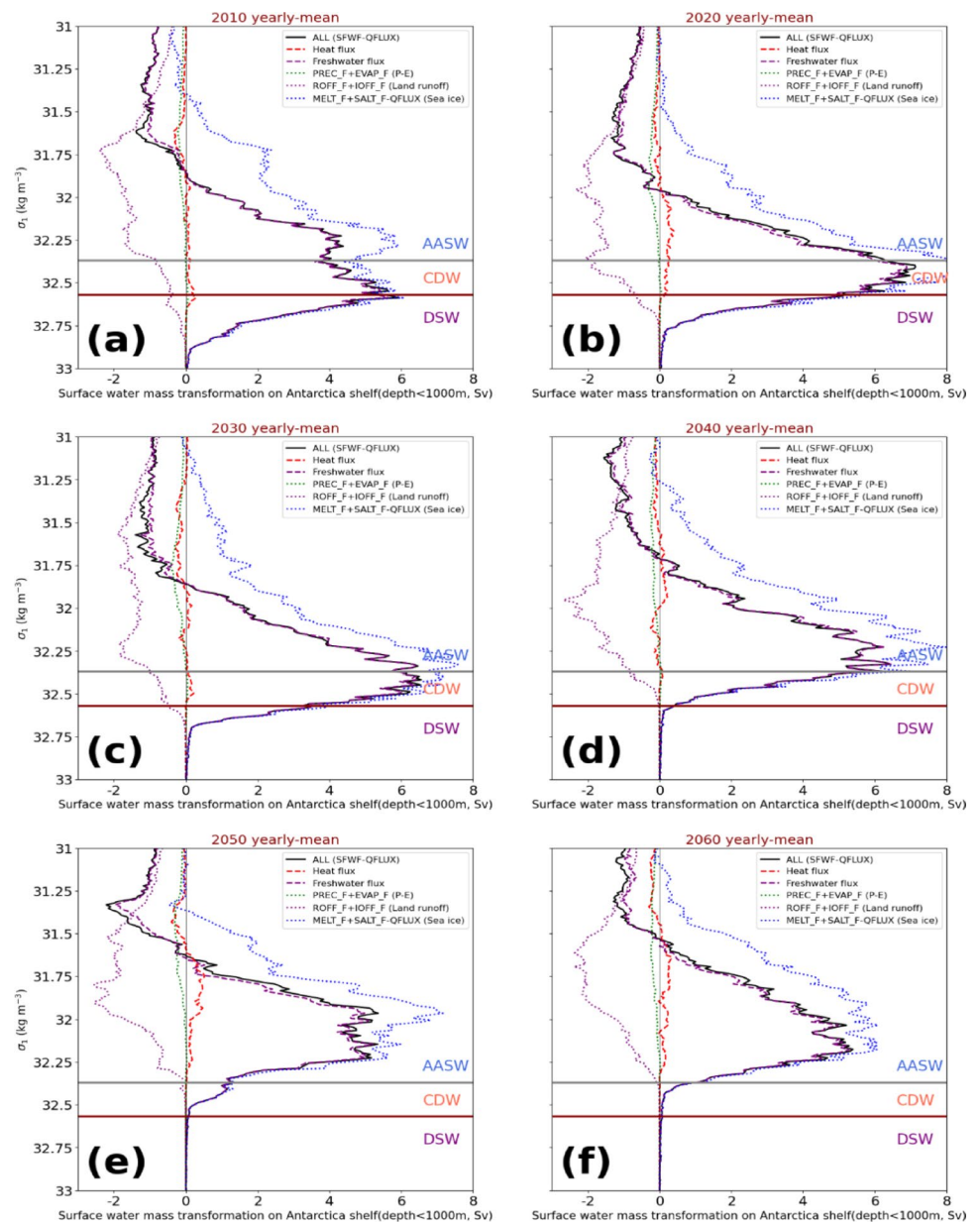


in winds and thermal forcing have little impacts on the DSW overflow in their simulations.

In this research, there is no a coupled dynamic ice sheet or ice shelf model in CESM1.3 (instead, CESM1.3 uses a snow runoff scheme to roughly represent this effect; refer to the Method section for detail). To investigate the processes that can potentially affect the salinity in the shelf waters, we thus decompose the surface freshwater flux into three components (Fig. 12): (1) induced by sea-ice brine rejection and melt, (2) induced by precipitation minus evaporation, (3) induced by snow/ice runoff and liquid water runoff

from the Antarctic continent. The results indicate that the change in the surface freshwater flux for the period of 2010–2100 is mostly dominated by the component induced by sea-ice brine rejection and melt on the Antarctic shelf (58%, for the slope, Fig. 12b). The second important term is the ice discharge and liquid water runoff from the Antarctic continent (26%, for the slope, Fig. 12c). The precipitation minus evaporation on the Antarctic shelf also makes a small contribution to the DSW shut down process (16%, for the slope, Fig. 12d). Further decomposition shows that snow/ice runoff contributes 80% of the total runoff, and liquid

Fig. 7 The surface water mass transformation is decomposed into the thermal and haline parts induced by heat flux and freshwater flux. Furthermore, the freshwater flux is decomposed into three components: precipitation minus evaporation on Antarctica shelf, snow/ice runoff and liquid water runoff from the Antarctic continent or ice sheet, and sea-ice brine rejection and melt on Antarctica shelf. **a–f** for year 2010, 2020, 2030, 2040, 2050, 2060, respectively. The horizontal lines and colored labels show the same water mass definitions as in Fig. 6 (the red line is 32.57 kg/m^3 , while the grey line is 32.37 kg/m^3)



water runoff is only accounting for 20% of the proportion around 2010 (Fig. 12e). However, under the climate forcing, the change of liquid water runoff is increasing faster than the snow/ice runoff change (1.630 vs 0.423 Sv/year for their slope in Fig. 12e). Furthermore, we also want to point out here that the snow/ice runoff into the ocean may not directly contributed to the sea water salinity change (as the liquid water runoff does), because when the land snow/ice melt, the heat exchange will immediately lead to the formation of sea-ice (frazil ice in the model) when sea water is near its freezing point. Compared with Li et al. (2023), their artificially increased liquid water runoff is from 0 to 0.08 Sv at a rate of 0.16Sv/century for the whole Antarctic shelf region, our simulated liquid water runoff is only increasing from

0.019Sv to about 0.03Sv. This may explain why the liquid water runoff is more dominate in the slowdown of the DSW overflow simulated by Li et al., (2023).

To check the spatial patterns of the changes in the total freshwater flux and its three components, we show their 20-year means over 1900–1920 and the differences between 2030–2050 and 1900–1920 (Fig. 13). The results also indicate that the total surface freshwater flux (and its change) is mainly dominated by the component induced by sea-ice brine rejection and melt on the Antarctic shelf (Fig. 13a, c and Fig. 13b, d). The snow/ice runoff and liquid water runoff from the Antarctic continent (and its change) plays a secondary role (Fig. 13e, f). Note that although both the snow/ice runoff and liquid water runoff appear to be directly

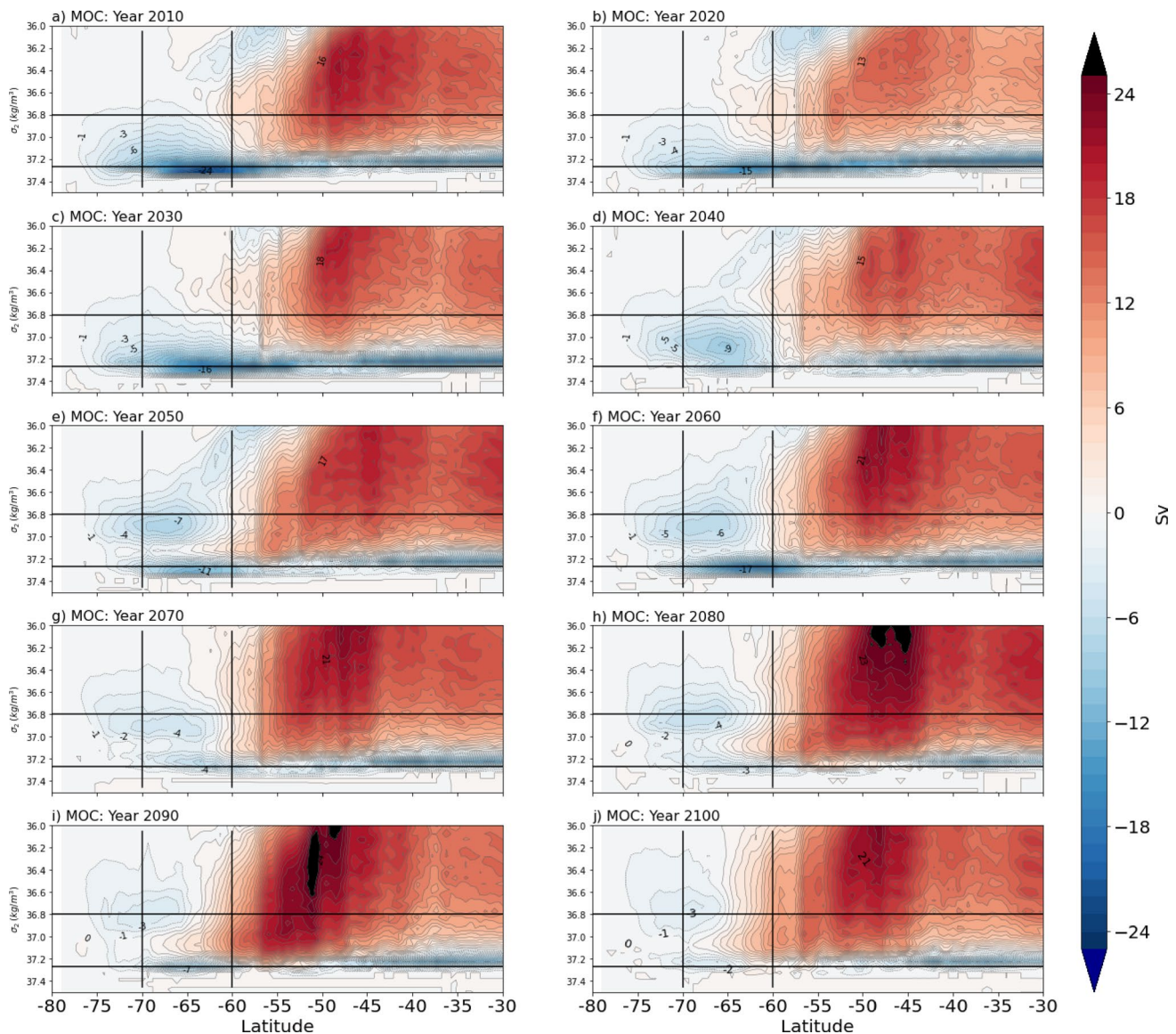


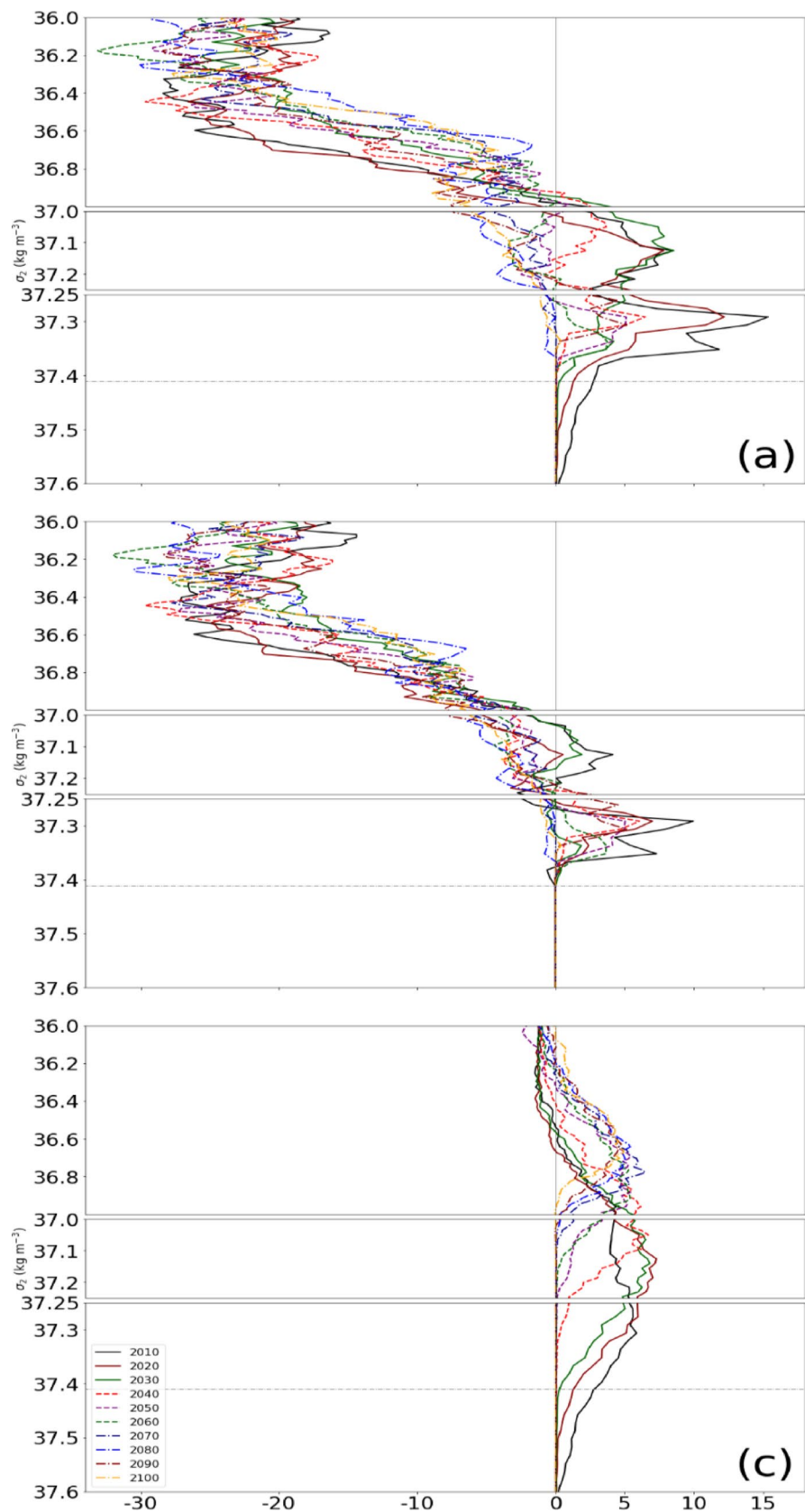
Fig. 8 Yearly-mean meridional overturning circulation (MOC). **a–j** year 2010, 2020, ..., 2100. The two horizontal black lines represent the $\sigma_2 = 37.25 \text{ kg/m}^3$ and $\sigma_2 = 36.8 \text{ kg/m}^3$, while the two vertical black lines represent 70°S and 60°S

released into the closest grids to the ice shelf fronts or coastline, much more runoff is released in summer than in winter (not shown), so its direct effect on the DSW formation in the coastal polynyas in winter is mitigated by the shelf sea circulation. Furthermore, the precipitation minus evaporation also increases, but generally very small over a broad region.

Our results thus highlight the importance of the brine rejection effect in driving the weak of DSW formation and export, in addition to the ice-sheet meltwater mechanisms as some previous studies have already proposed (e.g., Li et al. 2023; Moorman et al. 2020; Aguiar et al. 2023; Lago and England. 2019). However, the brine rejection effect is affected by several factors, such as the strength of offshore winds, the atmospheric and oceanic thermal conditions, and

the glacial meltwater from the Antarctic ice sheet (Stössel et al. 2011; Zhang et al. 2015; Jeong et al. 2023; Eayrs et al. 2021). It is likely that the sensitivities of these factors to climate change forcing are different in different models (Diamond et al. 2024; Roach et al. 2020). For example, the zonally averaged yearly mean 2-m air temperature in our simulations (Chang et al. 2020, their Fig. 26) is higher than the result in Li et al. (2023) (their extended Fig. 3). In addition, the imposed liquid freshwater runoff from ice-sheet meltwater in Li et al. (2023) (from 0 to 0.08 Sv) is much bigger than those simulated in this study (from 0.019Sv to about 0.03Sv, shown in the green line in Fig. 12e). These uncertainties, along with others, may inevitably lead to uncertain projections of future Antarctic Sea ice and ocean

Fig. 9 Yearly-mean surface water mass transformation **a** for all the Southern Ocean, 80°S–30°S. **b** for only the open-ocean part, with isobath > 1000 m. **c** for only the Antarctica shelf part, with isobath < 1000 m



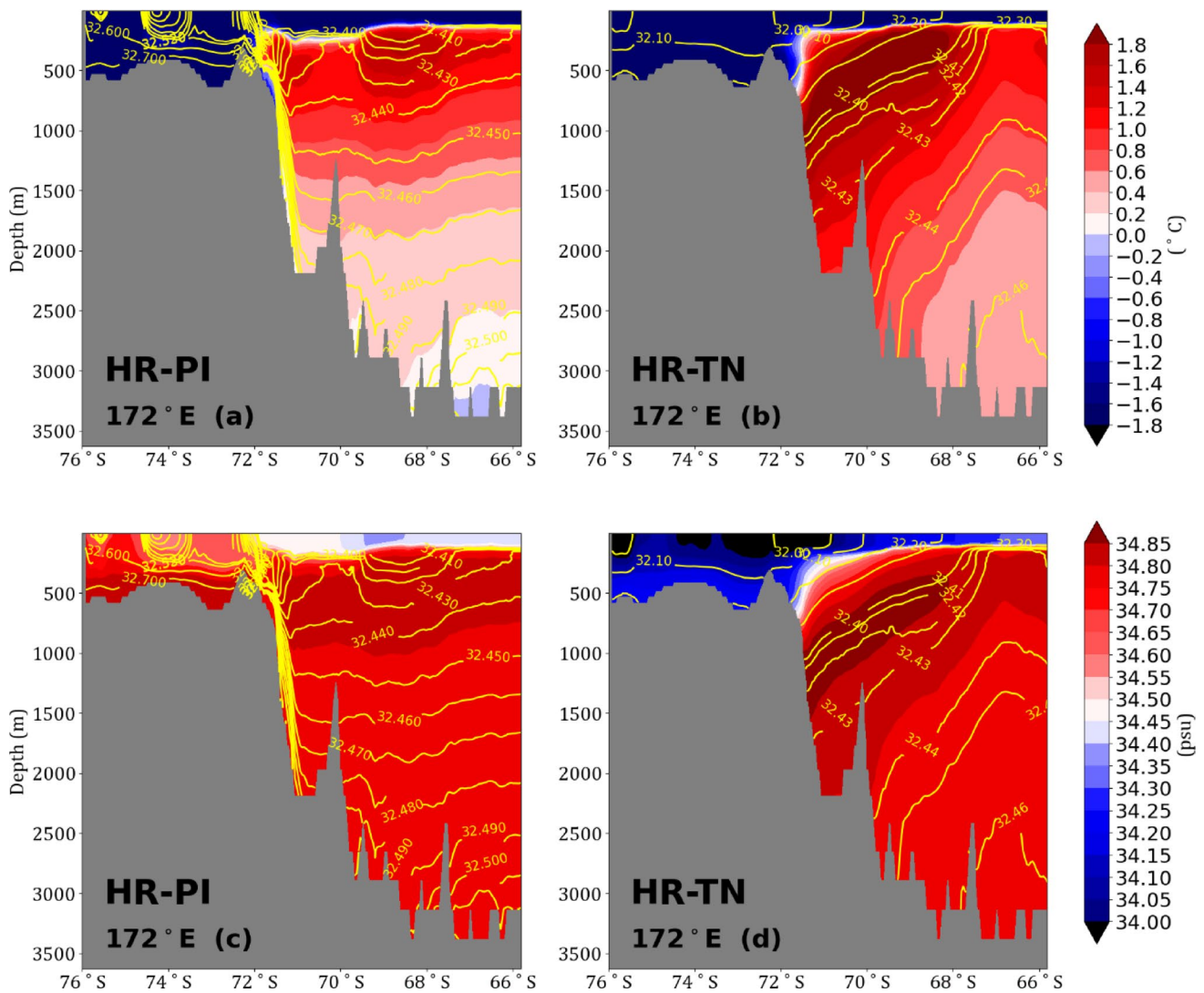


Fig. 10 Potential temperature (upper panels), salinity (lower panels), and potential density (yellow contour lines) in September along the Ross Sea section (172°E), time-averaged over the last 50 years of HR-PI (left panels) and of HR-TN (right panels)

circulation evolution in the CMIP6 models (Diamond et al. 2024; Roach et al. 2020; Nie et al. 2023). Our results derived from a particular model in this study thus warrant future studies for investigating how the freshwater forcing related to all major processes will respond to climate change forcing, to reduce uncertainties in future projections of DSW overflow.

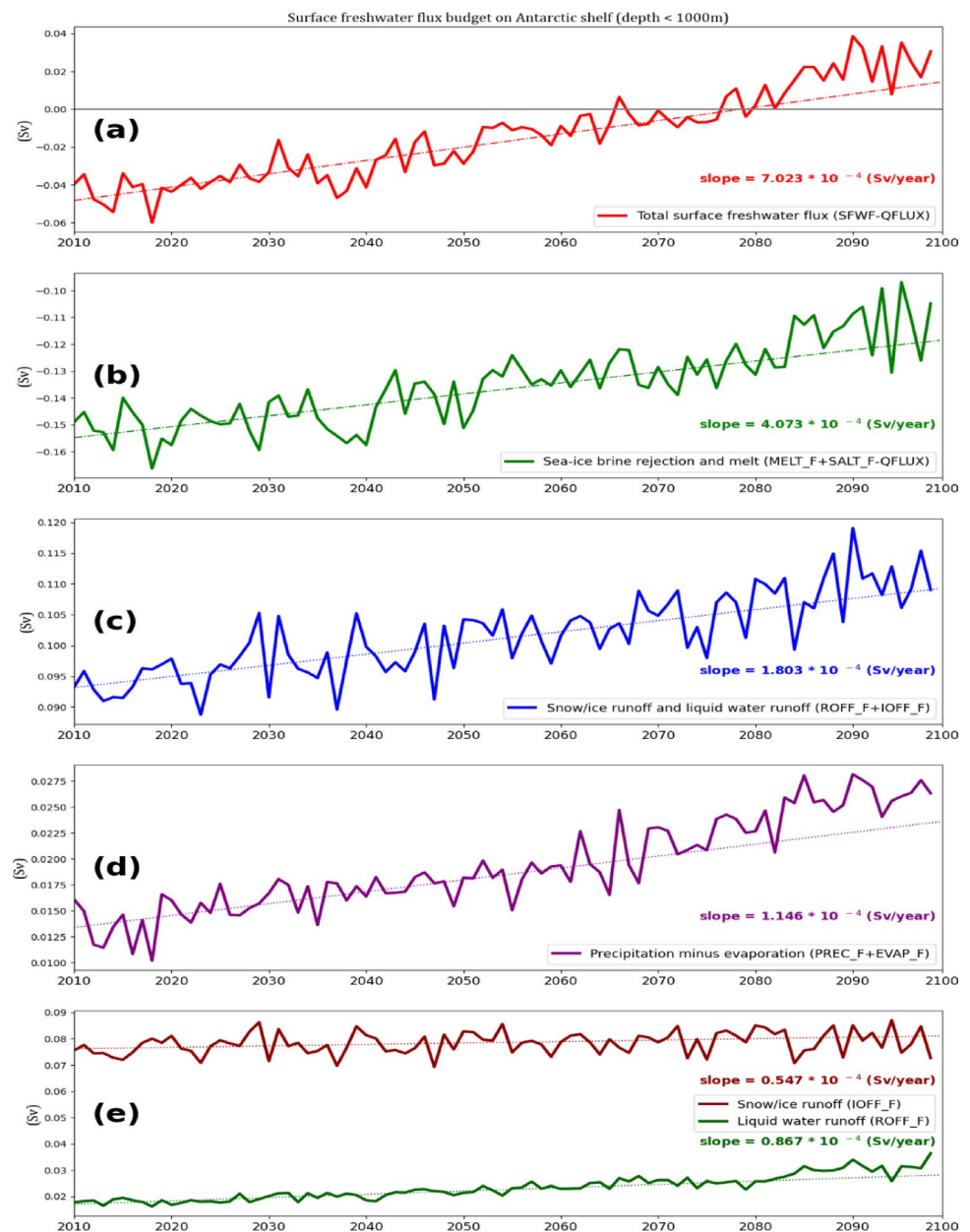
4 Summary and conclusions

This study investigates the effect of the anthropogenic climate change forcing on dense shelf water (DSW) overflow around Antarctica and the abyssal global ocean overturning circulation based on the fully coupled CESM1.3 model simulations. We have done a detailed total freshwater flux budget analysis on the Antarctica shelf in the CESM1.3

POP model. Our results highlight the importance of the brine rejection effect in driving the cessation of the DSW overflow in a high-emission scenario in this model. The following conclusions are drawn:

1. The high-resolution CESM 1.3 simulations captures the DSW overflow at the four locations where observations have confirmed the formation and export of DSW: the western Ross Sea shelf, the Adélie Coast, the western Weddell Sea shelf, and the Prydz Bay (Cape Darnley) region.
2. The anthropogenic climate forcing in high-resolution historical and future transient simulation (HR-TN) makes the DSW overflow come to a halt around 2040. The DSW overflow cessation is mainly due to the freshening of the Antarctic shelf water.

Fig. 12 a The total surface freshwater flux time-series (Sv) integrated over the whole Antarctica shelf (<1000 m isobath), and the three components: **b** sea-ice brine rejection and melt on the Antarctic shelf, **c** snow/ice runoff and liquid water runoff from the Antarctic continent, **d** precipitation minus evaporation on the Antarctic shelf. We also separated the snow/ice runoff and liquid water runoff in (e). Here we choose the period 2010–2060 to calculate their trends, which is sufficient to illustrate the long-term trend without including too many years after the DSW collapse



changes indicate that it is a challenge for the current-generation climate models to faithfully simulate the past and project the future variability and trend of the Antarctica sea-ice and the associated brine rejection effect. Also, a major caveat in most coupled climate models, including the high-resolution coupled CESM1.3 used in this study, is the absence of an active coupled ice sheet/shelf component (instead, CESM1.3 uses a freshwater runoff scheme to roughly represent this effect; see Sect. 2), which is likely one of the major uncertainty sources for the projected Antarctic sea-ice, freshwater forcing, and DSW overflow. Both observation and simulation results have indicated that the Antarctic ice sheet melt water has considerable impacts on Antarctic sea-ice and DSW overflow as it directly affects

the surface freshwater flux, sea ice circulation and water mass properties (Adusumilli et al. 2020; Eayrs et al. 2021; Schneider and Deser 2018; Li et al. 2023; Thomas et al. 2023). Furthermore, as the weakening of DSW overflow warms the subsurface ocean around Antarctica, this subsurface warming could further increase the basal melting of the ice shelves (Muntjewerf et al. 2021; Smith et al. 2021). In future studies, high-resolution fully-coupled climate models that include a coupled dynamic ice sheet component need to be developed to deliver more complete understanding and more reliable projections of the anthropogenic impact on DSW overflow.

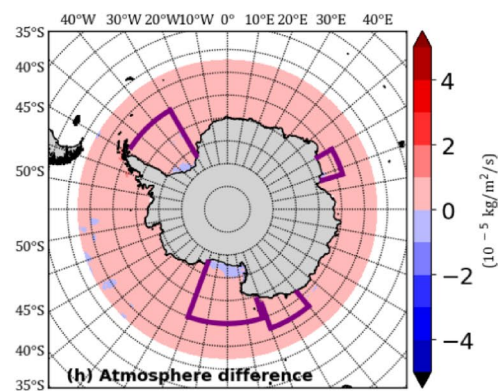
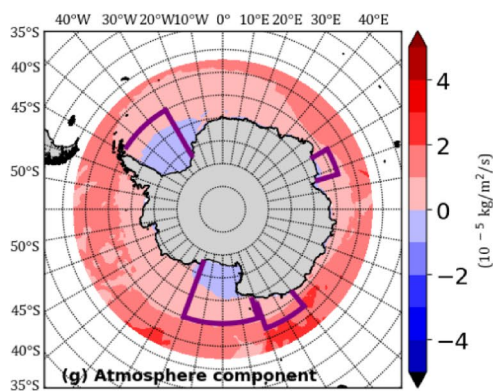
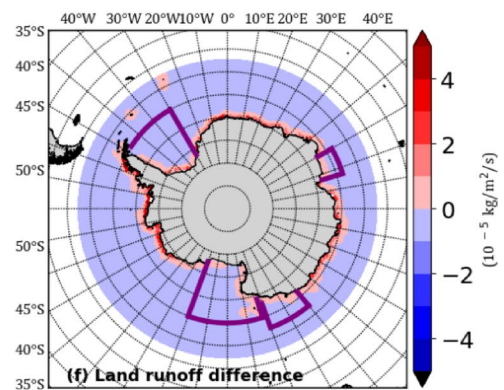
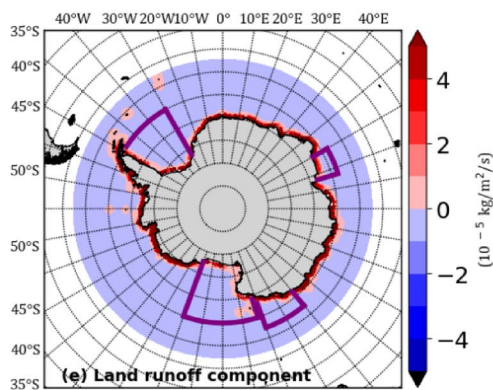
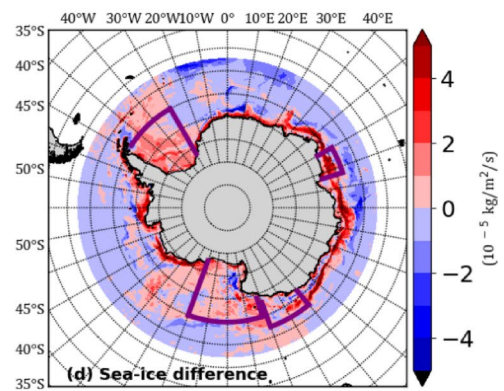
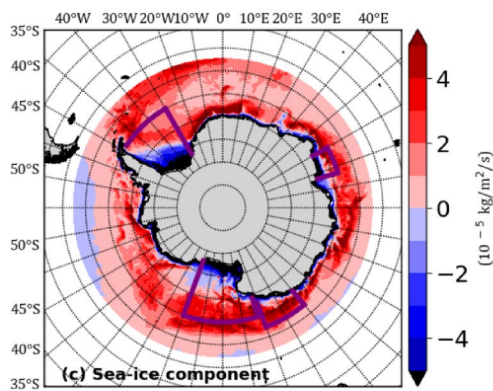
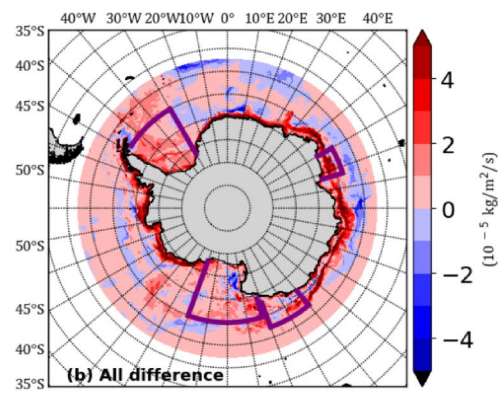
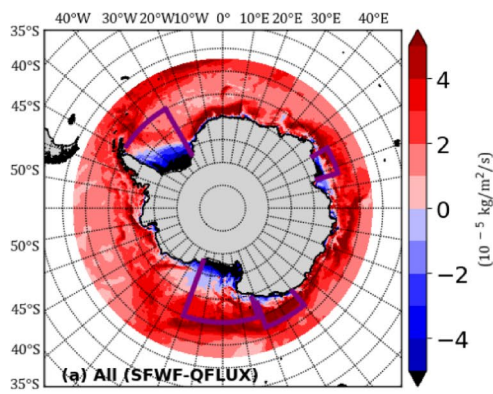


Fig. 13 a The 20-year averaged total surface freshwater flux over 1900–1920. **b** The total surface freshwater flux difference between 1900–1920 and 2030–2050. **(c)** and **(d)** Similar to **(a)** and **(b)**, but for the component induced by sea-ice brine rejection and melt. **(e)** and **(f)** Similar to **(a)** and **(b)**, but for the component induced by snow/ice runoff and liquid water runoff from the Antarctic continent. **(g)** and **(h)** Similar to **(a)** and **(b)**, but for the component induced by precipitation minus evaporation

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00382-025-07866-7>.

Acknowledgements This work was supported by the National Natural Science Foundation of China (NSFC; 42230405 and 41976006), the Laoshan Laboratory (No. LSKJ202202401), and the National Key R&D Program of China (2022YFE0106400). We also want to thank Achim Stössel, Ping Chang, Abishek Gopal, Sanjiv Ramachandran for their help with the CESM-iHESP datasets, simulations and valuable discussions here. Paul Spence is supported by Australian Research Council grants FT190100413, SR200100008, CE230100012, DP190100494, DP240102358, LP200100406. Xiliang Diao is supported by Shandong Province Natural Science Foundation Youth Branch (ZR2024QD142) and the Fundamental Research Funds for the Central Universities (202513020).

Data availability The data used in this work are available at the iHESP data portal (https://datahub.geos.tamu.edu:8880/thredds/catalog/iHESPDataHUB/B.E.13.B1850C5.ne120_t12.sehires38.003.sunway_02/catalog.html). All the calculation, analyzation, and plot codes can be found here (<https://github.com/xiliangdiao/Ross-Sea-DSW---project>).

Declarations

Competing interests The authors declare no competing interests.

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